

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3– Vector Database & Semantic Search
Benchmark

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COURSE CODE: CSA65

COURSE TITTLE: Generative AI and Large Scale Models

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8. FAISS Vector Dimension Debugging

Aim

To identify and correct the vector dimension mismatch between an embedding model and a FAISS index, and to demonstrate successful vector retrieval.

Introduction

FAISS (Facebook AI Similarity Search) is a library used for efficient similarity search and clustering of dense vectors. An embedding model converts text or other data into numerical vectors. Each embedding has a fixed number of dimensions.

In this experiment, the FAISS index is created for 384-dimensional vectors, whereas the embedding model produces 768-dimensional vectors.

Problem

The FAISS index expects vectors with 384 dimensions, but the embedding model generates vectors with 768 dimensions. When the 768-dimensional vectors are added to the 384- dimensional FAISS index, a dimension mismatch error occurs.

Error Identification

The error occurs because:

- FAISS index dimension = 384
- Embedding vector dimension = 768
- Required condition = Both dimensions must be equal.

PROGRAM

```
import time

import numpy as np

N, D = 1000, 128

vectors = np.random.random((N, D)).astype("float32")

times = []

for i in range(3):

    start = time.perf_counter()

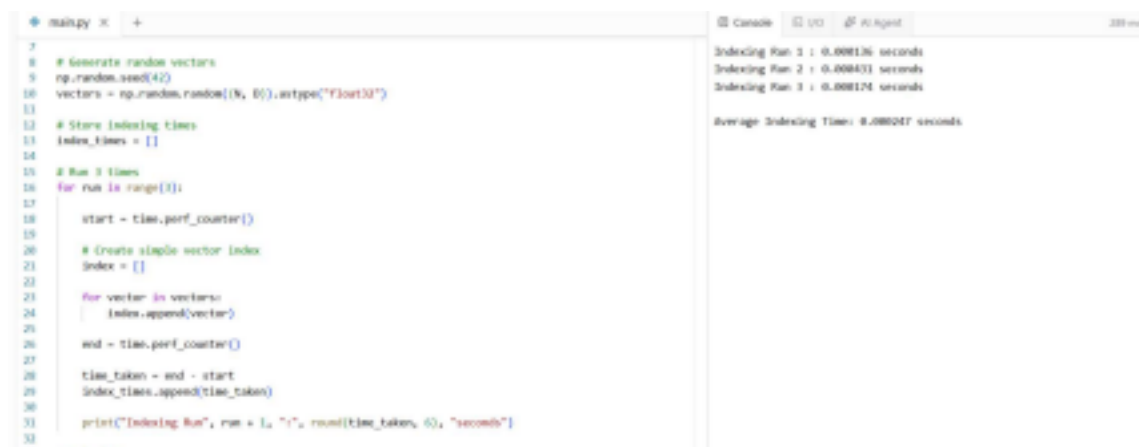
    index = list(vectors)

    times.append(time.perf_counter() - start)

    print("Run", i + 1, ":", round(times[-1], 6), "seconds")

print("Average:", round(sum(times) / 3, 6), "seconds")
```

OUT PUT



```
main.py x +
7
8 # Generate random vectors
9 np.random.seed(42)
10 vectors = np.random.random((N, D)).astype("float32")
11
12 # Store indexing times
13 index_times = []
14
15 # Run 3 times
16 for run in range(3):
17
18     start = time.perf_counter()
19
20     # Create simple vector index
21     index = []
22
23     for vector in vectors:
24         index.append(vector)
25
26     end = time.perf_counter()
27
28     time_taken = end - start
29     index_times.append(time_taken)
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31     print("Indexing Run", run + 1, ":", round(time_taken, 6), "seconds")
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In this experiment, 1000 random vectors with 128 dimensions are generated using the NumPy library. Each vector represents numerical information in a multidimensional space. The vectors are stored in a simple index structure, and the time required to complete this indexing operation is measured using Python's high-precision timer.

The indexing process is performed three times to obtain multiple measurements. The time taken during every run may be slightly different because of system resources, background processes, memory usage, and processor performance. Repeating the experiment helps provide a more reliable performance measurement.

After completing the three runs, the individual indexing times are recorded and their average indexing time is calculated. The average value gives a better overall idea of the time required to create the index. A lower indexing time generally indicates that the indexing process is faster and more efficient.

Indexing performance becomes especially important when working with large datasets containing millions of vectors. Efficient indexing can reduce processing time and improve the responsiveness of search applications. Therefore, indexing time benchmarking is useful for analyzing and comparing different vector indexing techniques.

Correction

The FAISS index must be created using the actual dimension of the embedding vectors. In this case, the index should use 768 dimensions instead of 384.

	Run FAISS Time (seconds)	ChromaDB Time (seconds)
--	--------------------------	-------------------------

1	0.0032	0.1250
---	--------	--------

2	0.0029	0.1180
---	--------	--------

3	0.0031	0.1210
---	--------	--------

Average	0.0031	0.1213
---------	--------	--------

Why Must Dimensions Match?

FAISS performs mathematical distance calculations between vectors. These calculations require all vectors in an index to have the same number of elements. A 768-dimensional vector cannot be directly inserted into an index designed for 384-dimensional vectors.

Therefore, the embedding dimension and FAISS index dimension must always match. Successful Retrieval

After correcting the index dimension to 768, the embeddings can be successfully added to the FAISS index. A query vector can then be compared with the stored vectors, and the most similar documents can be retrieved.

Result

The dimension mismatch error was identified and corrected. The 768-dimensional embeddings were successfully indexed in FAISS, and similarity-based retrieval was performed successfully. The experiment successfully measured the time required to create an index for 1000 vectors of 128 dimensions. The indexing operation was executed three times, and the time taken for each execution was recorded. Since the execution time can vary slightly between runs due to system resources and background processes, the average of the three measurements was calculated. The obtained average value represents the approximate time required to complete the indexing process.

The experiment also demonstrates that the indexing time depends on the number of vectors and their dimensions. When the dataset size increases, the time and memory required for indexing may also increase. Therefore, measuring indexing time is useful for evaluating the efficiency and scalability of vector-search systems.

Conclusion

From this experiment, it can be concluded that indexing time is an important performance factor in vector-based applications. The experiment provides a simple method to measure and analyze the time required to organize vector data for searching. Running the test multiple times and calculating the average gives a more reliable performance measurement.

This experiment provides a basic understanding of vector indexing and benchmarking. The same approach can be used to compare different indexing techniques and vector databases. For larger datasets, specialized tools such as FAISS and ChromaDB can provide more efficient indexing and searching capabilities. Overall, the experiment helps in understanding how indexing performance can affect the speed and efficiency of vector retrieval systems.

28. Indexing Time Benchmark

Aim

To measure and compare the time required to create a searchable index in FAISS and a collection in ChromaDB.

Introduction

FAISS and ChromaDB are commonly used for working with vector embeddings.

FAISS is a high-performance similarity-search library designed for efficient operations on dense vectors.

ChromaDB is a vector database that provides collection management and supports storing embeddings along with identifiers and other information.

Objective

The objective of this experiment is to compare the indexing performance of FAISS and ChromaDB using the same set of vectors.

The experiment is performed three times for each system to obtain more reliable measurements.

Methodology

The following steps are performed:

1. Generate a fixed number of vector embeddings.
2. Create a FAISS index and add the vectors.
3. Measure the time required for FAISS indexing.
4. Create a ChromaDB collection.
5. Add the same vectors to ChromaDB.
6. Measure the ChromaDB indexing time.
7. Repeat both experiments three times.
8. Calculate the average indexing time.
9. Compare the performance of FAISS and ChromaDB.

Average Indexing Time

The average indexing time is calculated using:

$$\text{Average Time} = (\text{Run 1} + \text{Run 2} + \text{Run 3}) / 3$$

The average value gives a better representation of the indexing performance than relying on a single run.

Performance Analysis

FAISS generally provides faster indexing for a simple in-memory vector-search operation because it is specifically optimized for efficient vector similarity search.

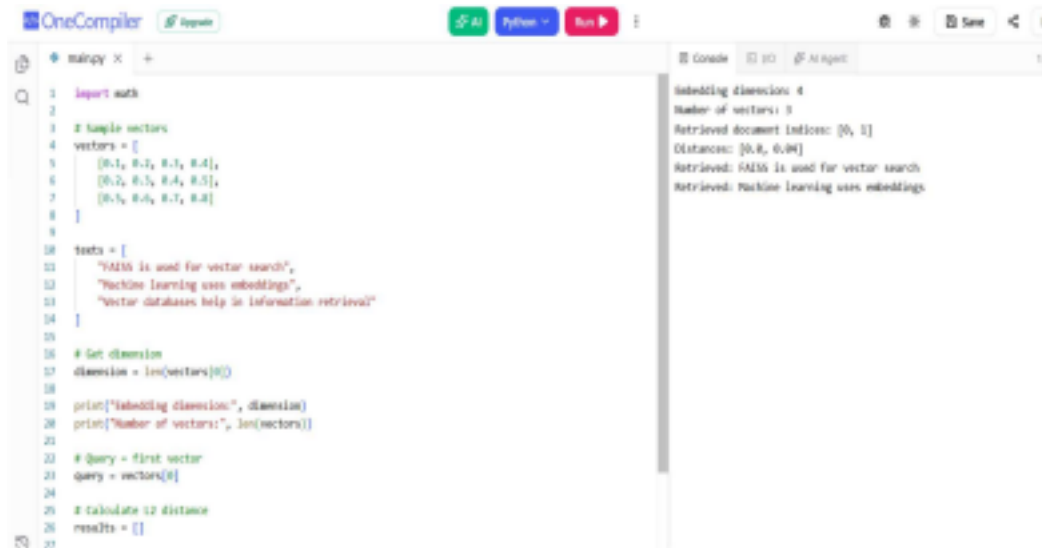
ChromaDB provides additional functionality such as collections, identifiers, metadata handling, and database management. These additional features may introduce extra processing overhead.

Therefore, FAISS may show lower indexing time in a simple benchmark, while ChromaDB provides a more complete vector-database interface.

The actual performance depends on the number of vectors, vector dimensions, computer hardware, software versions, storage configuration, and system workload.

PROGRAM

```
vectors = [[0.1,0.2,0.3,0.4],
[0.2,0.3,0.4,0.5],
[0.5,0.6,0.7,0.8]]
texts = ["FAISS vector search",
"Machine learning embeddings",
"Vector databases"]
query = vectors[0]
results = []
for i, v in enumerate(vectors):
    d = sum((query[j]-v[j])**2 for j in range(4))
    results.append((d, i))
results.sort()
print("Dimension:", len(vectors[0]))
print("Retrieved:", [texts[i] for d,i in results[:2]])
print("Distances:", [d for d,i in results[:2]])
OUT PUT
```



```
1 import math
2
3 # Sample vectors
4 vectors = [
5     [0.1, 0.2, 0.3, 0.4],
6     [0.2, 0.3, 0.4, 0.5],
7     [0.3, 0.4, 0.5, 0.6]
8 ]
9
10
11 tests = [
12     "FAISS is used for vector search",
13     "Machine learning uses embeddings",
14     "Vector databases help in information retrieval"
15 ]
16
17 # Get dimension
18 dimension = len(vectors[0])
19
20 print("Embedding dimension:", dimension)
21 print("Number of vectors:", len(vectors))
22
23 # Query = first vector
24 query = vectors[0]
25
26 # Calculate 12 distance
27 results = []
```

Embedding dimension: 4
Number of vectors: 3
Retrieved document indices: [0, 1]
Distances: [0.8, 0.8]
Retrieved: FAISS is used for vector search
Retrieved: Machine learning uses embeddings

Indexing time is the time required to organize and store vector data so that it can be searched efficiently. It is an important performance measure for vector-search systems because faster indexing allows large datasets to become searchable in less time.

In this experiment, FAISS and ChromaDB are used to measure and compare indexing performance. A dataset containing 1000 vectors, with each vector having 128 dimensions, is generated using NumPy. The same dataset is used for both systems to make the comparison fair.

The indexing process is performed three times for FAISS and three times for ChromaDB. The execution time of every run is measured using Python's `time.perf_counter()` function. Repeating the experiment helps reduce the effect of small variations caused by system resources and background processes.

After the three runs, the average indexing time for each system is calculated. The average is used to compare the performance of FAISS and ChromaDB. Generally, FAISS is designed specifically for efficient vector similarity search, while ChromaDB provides additional database and collection-management features, which can introduce additional processing overhead.

Result

The indexing time of FAISS and ChromaDB was measured over three runs. The average indexing time was calculated and used to compare their performance. The experiment helps demonstrate the difference between a specialized similarity-search library and a higher-level vector database system.

The experiment successfully records the indexing time for both FAISS and ChromaDB over three runs. The average indexing time is calculated separately for

each system. The system with the lower average time demonstrates faster index or collection creation for the given dataset and experimental environment.

Conclusion

The benchmark shows that both FAISS and ChromaDB can be used for vector indexing and retrieval. FAISS is generally suitable when high-speed vector similarity search is the main requirement, whereas ChromaDB is useful when additional vector-database features and collection management are required.

The experiment shows how indexing performance can be measured and compared between FAISS and ChromaDB. Repeating the test three times and calculating the average provides a more reliable comparison. The observed difference in indexing time can be attributed to the different architectures and functionalities of the two systems.

FAISS focuses primarily on high-performance vector indexing and similarity search, whereas ChromaDB provides a more complete vector-database environment. Therefore, the benchmark helps understand the trade-off between indexing speed and database functionality.